

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2021-22

Paper: BVCA-205 — Skill Training in Python

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt all questions. Part A: Multiple Choice and Short Questions. Part B: Internal Choice (any four). Part C: Open choice (any three out of four).

PART — A*Note: Multiple Choice Type Questions and Short Questions.***[10 × 2 = 20]****Question 1.** Which of the following is a Python tuple?

- (a) {1, 2, 3}
- (b) 4
- (c) [1, 2, 3]
- (d) (2, 2, 3)

Question 2. Which one of the following is not a keyword in Python?

- (a) `pass`
- (b) `eval`
- (c) `assert`
- (d) `nonlocal`

Question 3. In object-oriented programming, what is the relation of an object to a class?

- (a) Class is an instance of an object.
- (b) Object is a child of a class.
- (c) Object is a method of a class.
- (d) Object is an instance of a class.

Question 4. What will be the output of the following Python code snippet?

```
print(type(5/2))  
print(type(5//2))
```

- (a) float and int
- (b) float and float

- (c) int and int
- (d) int and float

Question 5. What will be the output of the following Python code snippet?

```
print(2**4 + (5+5)**(1+1))
```

- (a) Error
- (b) 28
- (c) 118
- (d) 116

Question 6. What will be the output of the following Python code?

```
i = 1
while True:
    if i % 3 == 0:
        break
    print(i)
    i += 1
```

- (a) 1 2 3
- (b) Error
- (c) 1 2
- (d) None of the mentioned

Question 7. Which one of the following has the same precedence level?

- (a) Addition and Subtraction
- (b) Multiplication, Division and Addition
- (c) Multiplication, Division, Addition and Subtraction
- (d) Addition and Multiplication

Question 8. What will be the datatype of `test` in the below code snippet?

```
test = 10
print(type(test))
test = "testbook"
print(type(test))
```

- (a) str and int
- (b) int and int
- (c) str and str

(d) int and str

Question 9. What is the meaning of `>>` in Python?

- (a) Interpreter is ready to take instruction.
- (b) Compiler is ready to take instruction.
- (c) Three right shift.
- (d) Three left shift.

Question 10. What is the output of `print(math.pow(3, 2))`?

- (a) 9.0
- (b) None
- (c) 9
- (d) None of the mentioned

PART — B

Note: Internal Choice — Either/Or type questions. Each question carries 5 marks. Attempt any four. **[5 × 4 = 20]**

Question 11. What are the common built-in data types in Python?
How do you read and write into Python files? Explain with a program.

Question 12. Explain the concept of overriding methods with a program.
Write a Python program to create a lambda function that returns the square value of a given number.

Question 13. Write a Python program to create a static method that calculates the square root value of a given number.

OR

Explain the `values()` and `items()` methods used in dictionaries with examples.

Question 14. What is the difference between Python Arrays, lists, and tuples with proper examples?

OR

Write a Python program to define a Student class and create an object. Also, display the student's details.

Question 15. What are the key features of Python over other programming languages?

OR

Write a function that accepts two values and finds their sum.

PART — C

Note: Open choice — Attempt any 3 out of 4 questions. Each question carries 10 marks. **[3 × 10 = 30]**

Question 16. List the operators that Python supports. Explain the relational and logical operators along with their precedence while evaluating an expression.

Question 17. Write short notes on:

- (a) Debugging
- (b) Exception Handling

Question 18. Write a Python program to retrieve the elements from a 2D array and display them using two different loops.

Question 19. Discuss the technical strengths of Python. What is the difference between list and tuples in Python?